

STAT/MA 41600

In-Class Problem Set #3: August 31, 2026

1. In a recent study that focuses only on the top 3 generative AI assistants, it is estimated that 50% of users prefer ChatGPT, 33% prefer Gemini, and 17% prefer Claude.

Consider a survey of five people, whose responses are assumed to be independent.

1a. What is the probability that none of them prefer Claude?

1b. What is the probability that all five of them agree on the same preferred assistant?

2. Roll a green 6-sided die and a red 6-sided die repeatedly. If the sum on the pair of dice is between 4 and 8 (inclusive), we call it a good result.

2a. What is the probability that the first 3 rolls of the pair of dice are all good results?

2b. Suppose that getting a sum of eleven is a bad result. What is the probability that we get a bad result before we get a good result?

3. Consider a randomly chosen bear family, consisting of 3 bears, namely, a daddy bear, a momma bear, and a baby bear. Each bear has six possible colors, namely, red, orange, yellow, green, blue, purple. If we pick such a bear family randomly, the sample space contains all ordered 3-tuples whose elements are r, o, y, g, b, p . Assume that all such outcomes are equally likely.

3a. What is the probability that a randomly selected bear family has at least one green bear?

3b. What is the probability that a randomly selected bear family has at least one green bear or at least one purple bear (or both)?

4. During a 5-day week (Monday to Friday), each day is sunny, cloudy, or rainy. If we pick a week randomly, the sample space contains all ordered 5-tuples whose elements are s, c , or r . Assume that all of the weekly outcomes are equally likely, and therefore, the weather is independent from day to day.

4a. Bob feels sad if, during the week, there are four or more rainy days. What is the probability that Bob feels sad about the weekly weather forecast?

4b. Mary feels sad if, during the week, there are three or more rainy days. What is the probability that Mary feels sad about the weekly weather forecast?

5. Bob, Catherine, Doug, and Edna are randomly assigned seats at a 4-sided table. All of the seating arrangements at the table are equally likely, because the waiter did not know how they are related.

Bob and Catherine are married. Doug and Edna are married. When people are married, they love to sit beside each other.

Let V denote the event that Bob and Catherine are sitting next to each other. Let W be the event that Doug and Edna are sitting next to each other.

Are events V and W independent or dependent? Why? Justify your answer.

(This is related to practice problem 3 from the problems posted on the internet, but it is different.)